

Effects of kiwifruit consumption on serum lipid profiles and antioxidative status in hyperlipidemic subjects.

Cardiovascular disease (CVD) is the most important adult health problem in the world. Epidemiological studies and laboratory experiments have shown that fruit and vegetable consumption has protective effects against CVD. The purpose of the study was to investigate the effects of consumption of two kiwifruit per day on the lipid profile, antioxidants and markers of lipid peroxidation in hyperlipidemic adult men and women in Taiwan. Forty-three subjects who had hyperlipidemia, including 13 males and 30 females, participated in this study. They were asked to consume two kiwifruit per day for 8 weeks. Anthropometric measurements were made. Before the intervention and at 4 and 8 weeks of the intervention, fasting blood samples were analyzed for total cholesterol, triacylglycerol, low-density lipoprotein (LDL) cholesterol, and high-density lipoprotein cholesterol (HDL-C). Additionally vitamin E and vitamin C, the malondialdehyde + 4-hydroxy-2(E)-nonenal concentration, and the lag time of LDL oxidation were determined. No significant differences from baseline to 8 weeks of the intervention were detected for triacylglycerol, total cholesterol, or LDL cholesterol. However, after 8 weeks of consumption of kiwifruit, the HDL-C concentration was significantly increased and the LDL cholesterol/HDL-C ratio and total cholesterol/HDL-C ratio were significantly decreased. Vitamin C and vitamin E also increased significantly. In addition, the lag time of LDL oxidation and malondialdehyde + 4-hydroxy-2(E)-nonenal had significantly changed at 4 and 8 weeks during the kiwifruit intervention. Regular consumption of kiwifruit might exert beneficial effects on the antioxidative status and the risk factors for CVD in hyperlipidemic subjects.